gocachemark - Go Cache Benchmark Report @ 2026-01-14 11:35:35 EST

gocachemark · darwin/arm64 · 16 CPUs · go1.25.5

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2026-01-14 11:35:35 EST gocachemark
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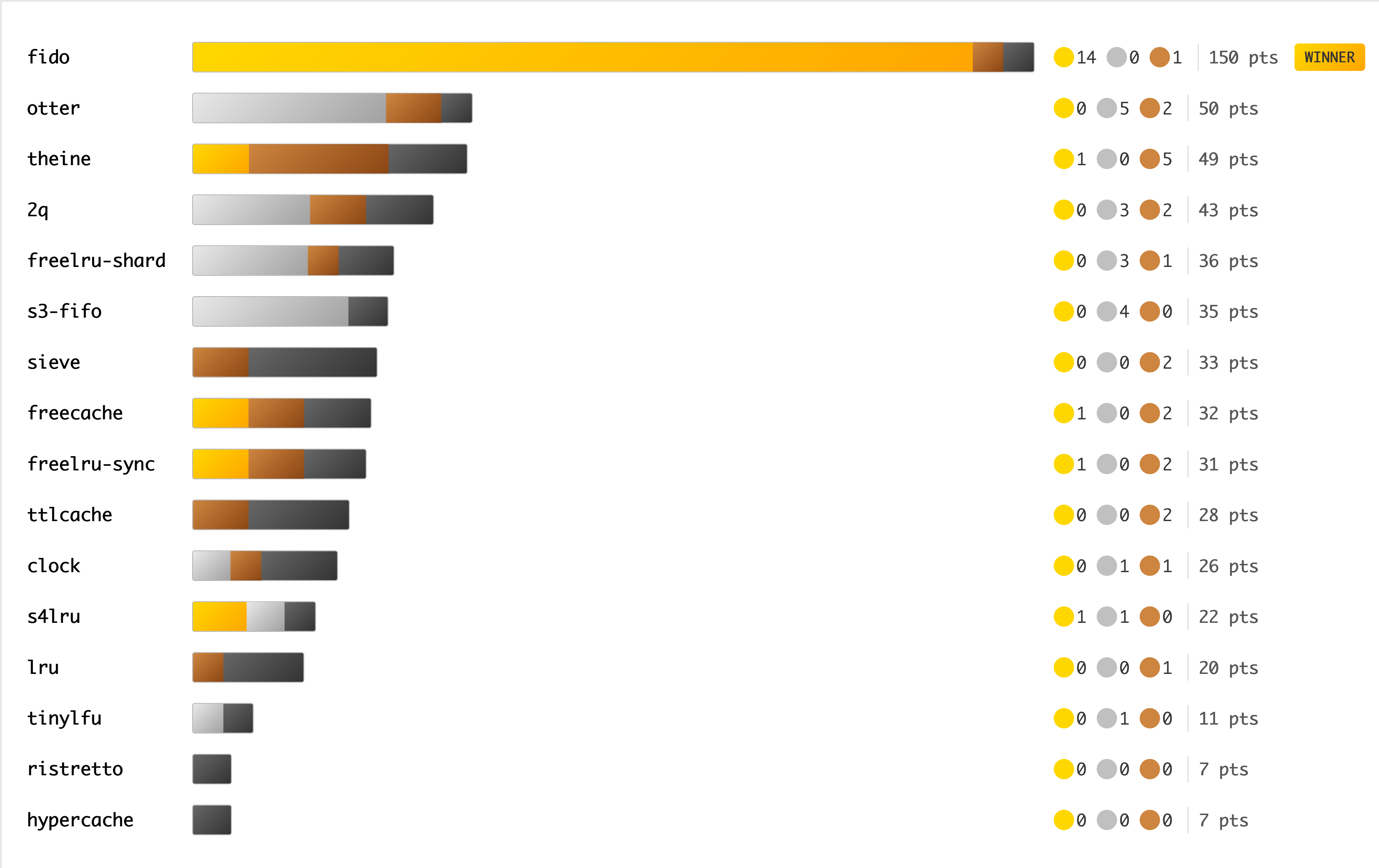
Medal Standings

Ranked by total points earned across all benchmarks



Medals by Category

Hit Rate — Winner: `fido`

fido	🟡 6 🟤 1
s4lru	🟡 1 ⚪ 1
theine	🟡 1 🟤 2
freecache	🟡 1
s3-fifo	⚪ 4
2q	⚪ 3 🟤 2
tinylfu	⚪ 1
sieve	🟤 2
otter	🟤 1
freelru-sync	🟤 1
ttlcache	🟤 1
lru	🟤 1
clock	🟤 1

Benchmark	Gold	Silver	Bronze
CDN	fido	s3-fifo	2q
Meta	fido	s3-fifo	theine
Zipf	theine	tinylfu	otter, fido
Twitter	fido	2q	sieve
Wikipedia	fido	s3-fifo	theine
Thesios Block	s4lru	2q	sieve
Thesios File	fido	s4lru	lru, ttlcache, freelru-sync
IBM Docker	freecache	2q	clock
Tencent Photo	fido	s3-fifo	2q

Latency — Winner: fido

fido	🟡 3
otter	⚪ 1 🟤 1
clock	⚪ 1
freelru-shard	⚪ 1
freecache	🟤 1
freelru-sync	🟤 1

Benchmark	Gold	Silver	Bronze
String Keys	fido	clock	freelru-sync
Int Keys	fido	freelru-shard	otter
GetOrSet	fido	otter	freecache

Throughput — Winner: fido

fido	🟡 5
otter	⚪ 3
freelru-shard	⚪ 2
theine	🟠 3
freecache	🟠 1
ttlcache	🟠 1

Benchmark	Gold	Silver	Bronze
String Get	fido	otter	theine
String Set	fido	freelru-shard	freecache
Int Get	fido	otter	theine
Int Set	fido	freelru-shard	ttlcache
GetOrSet	fido	otter	theine

Memory — Winner: `freelru-sync`

freelru-sync	🟡 1
otter	⚪ 1
freelru-shard	🟠 1

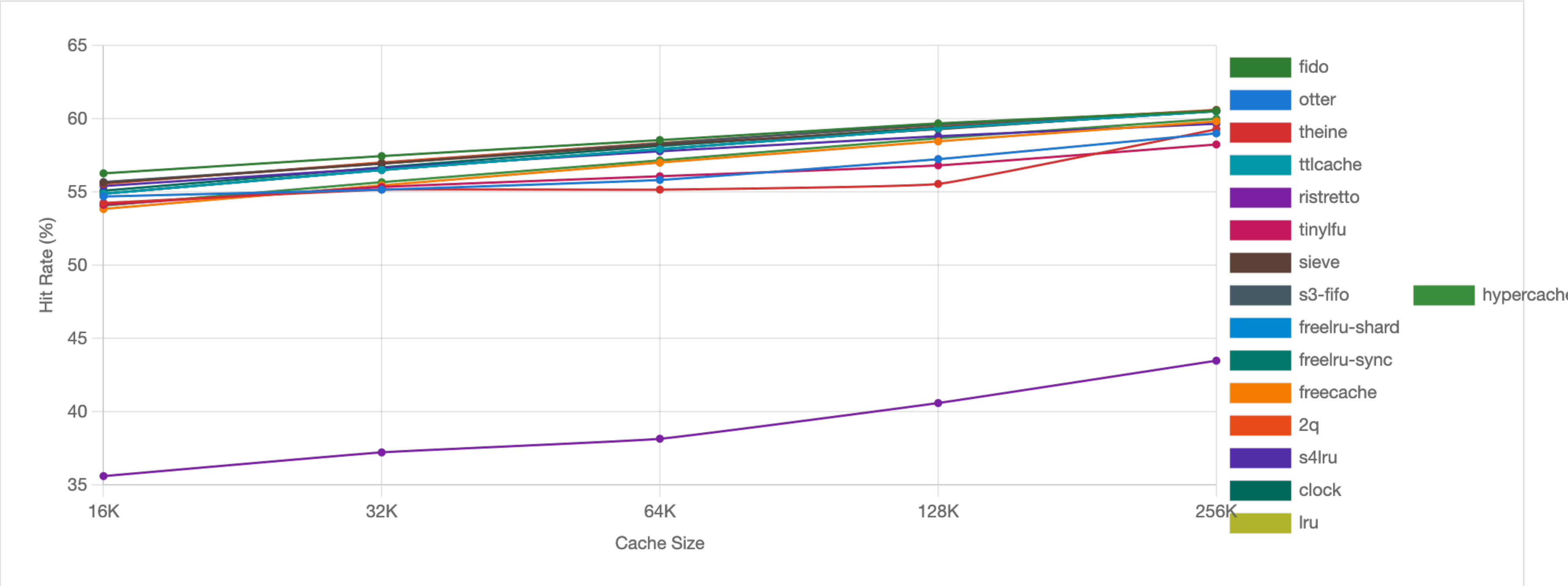
Benchmark	Gold	Silver	Bronze
Overhead	freelru-sync	otter	freelru-shard

Hit Rate Benchmarks ▼

> Higher hit rates mean better cache effectiveness. Tests measure how often requested items are found in the cache across different cache sizes.

CDN Production Trace

> 2M operations, ~768K unique keys. Source: [libCacheSim]



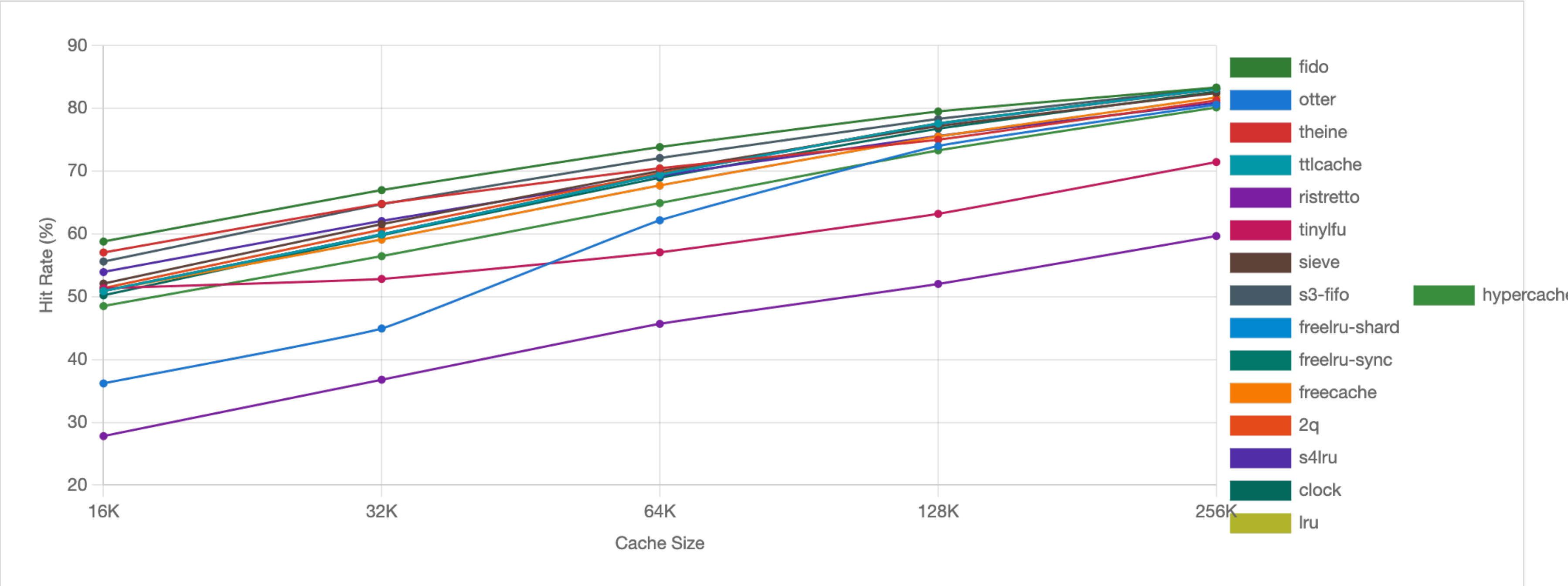
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	56.282%	57.457%	58.553%	59.694%	60.554%	58.508%
s3-fifo	55.690%	56.948%	58.327%	59.611%	60.554%	58.226%
2q	55.575%	57.019%	58.349%	59.559%	60.613%	58.223%
sieve	55.634%	56.947%	58.248%	59.298%	60.587%	58.143%
clock	55.093%	56.675%	58.138%	59.422%	60.566%	57.979%
freelru-sync	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%
lru	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%
ttlcache	54.901%	56.516%	57.944%	59.345%	60.533%	57.848%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	54.886%	56.507%	57.940%	59.344%	60.532%	57.842%
s4lru	55.423%	56.644%	57.789%	58.820%	59.674%	57.670%
hypercache	54.083%	55.685%	57.168%	58.675%	60.011%	57.124%
freecache	53.849%	55.457%	57.019%	58.468%	59.833%	56.925%
otter	54.704%	55.166%	55.829%	57.253%	59.013%	56.393%
tinylfu	54.139%	55.350%	56.081%	56.825%	58.248%	56.128%
theine	54.265%	55.166%	55.168%	55.559%	59.289%	55.889%
ristretto	35.609%	37.226%	38.157%	40.600%	43.490%	39.016%

Meta KVCache Production Trace

> 5M operations. Source: [CacheLib]



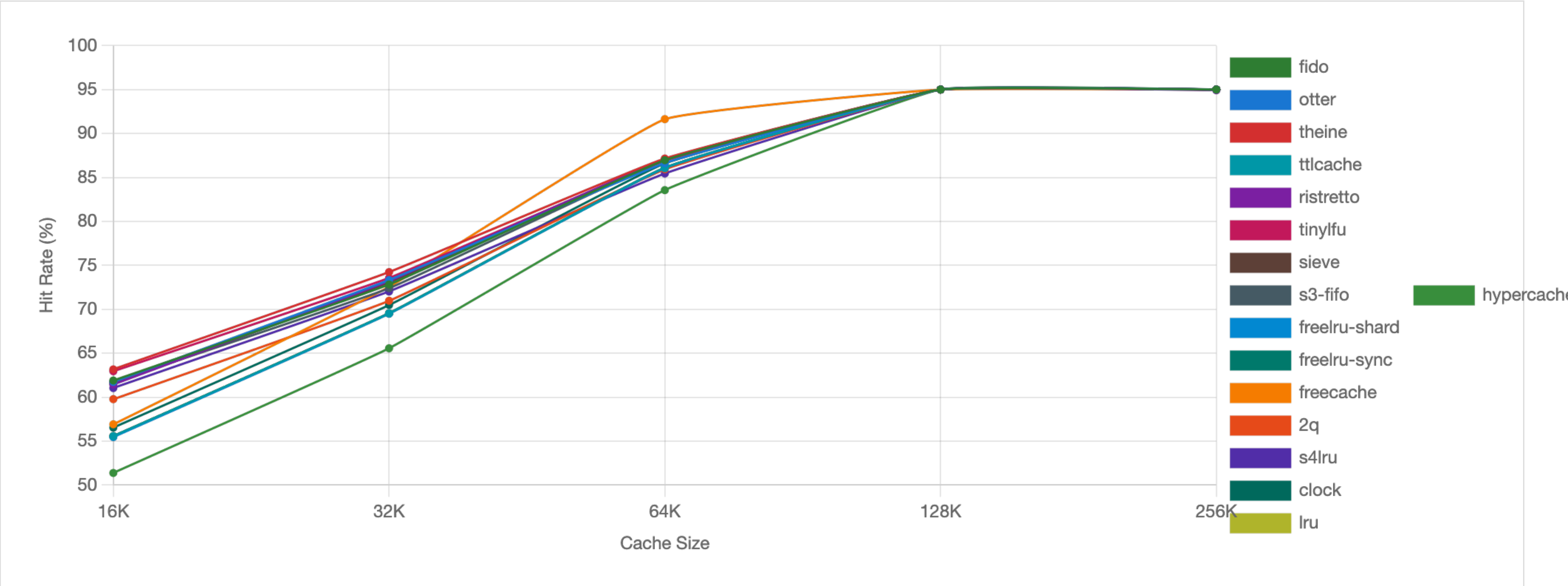
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	58.798%	66.991%	73.854%	79.505%	83.330%	72.495%
s3-fifo	55.609%	64.742%	72.101%	78.334%	83.275%	70.812%
theine	57.054%	64.805%	70.473%	75.016%	81.269%	69.724%
sieve	52.098%	61.564%	70.016%	77.166%	82.451%	68.659%
2q	51.375%	60.714%	69.647%	77.533%	83.073%	68.469%
s4lru	53.943%	62.068%	69.274%	75.591%	80.867%	68.349%
freelru-sync	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%
lru	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%

Cache	16K	32K	64K	128K	256K	Avg
ttlcache	50.934%	59.924%	69.436%	77.579%	83.116%	68.198%
freelru-shard	50.895%	59.941%	69.400%	77.509%	83.114%	68.172%
clock	50.228%	59.802%	68.951%	76.758%	82.677%	67.683%
freecache	50.926%	59.120%	67.731%	75.525%	81.733%	67.007%
hypercache	48.525%	56.469%	64.921%	73.303%	80.145%	64.672%
otter	36.203%	44.930%	62.170%	73.995%	80.576%	59.575%
tinylfu	51.354%	52.830%	57.071%	63.202%	71.448%	59.181%
ristretto	27.811%	36.779%	45.684%	52.032%	59.672%	44.396%

Zipf Synthetic Trace

> alpha=0.8, 2M operations, 100K keyspace



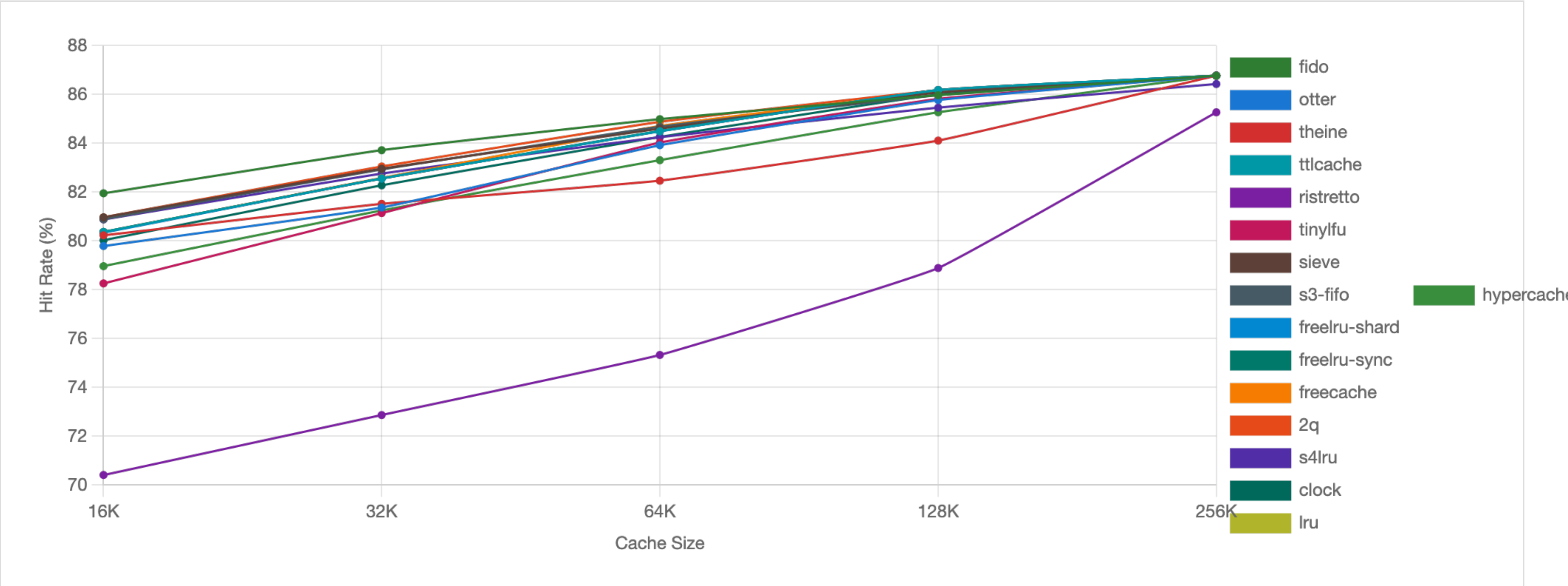
Results Table

Cache	16K	32K	64K	128K	256K	Avg
theine	63.174%	74.228%	87.148%	95.014%	95.014%	82.916%
tinylfu	62.964%	73.573%	86.917%	95.014%	95.014%	82.696%
otter	61.733%	73.331%	86.630%	95.014%	95.014%	82.344%
fido	61.896%	72.833%	86.965%	95.014%	95.014%	82.344%
ristretto	61.437%	73.213%	87.069%	95.008%	94.931%	82.332%
sieve	61.558%	73.071%	86.998%	95.014%	95.014%	82.331%
freecache	56.915%	72.559%	91.628%	95.014%	95.014%	82.226%
s3-fifo	61.740%	72.401%	86.912%	95.014%	95.014%	82.216%

Cache	16K	32K	64K	128K	256K	Avg
s4lru	61.043%	72.030%	85.442%	95.014%	95.014%	81.709%
2q	59.772%	70.969%	85.935%	95.014%	95.014%	81.341%
clock	56.541%	70.476%	86.600%	95.014%	95.014%	80.729%
freelru-sync	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
ttlcache	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
lru	55.569%	69.547%	86.112%	95.014%	95.014%	80.251%
freelru-shard	55.472%	69.499%	86.069%	95.014%	95.014%	80.214%
hypercache	51.380%	65.560%	83.560%	95.014%	95.014%	78.106%

Twitter Production Cache Trace

> 2M operations, cluster001+cluster052. Source: Twitter Cache Trace



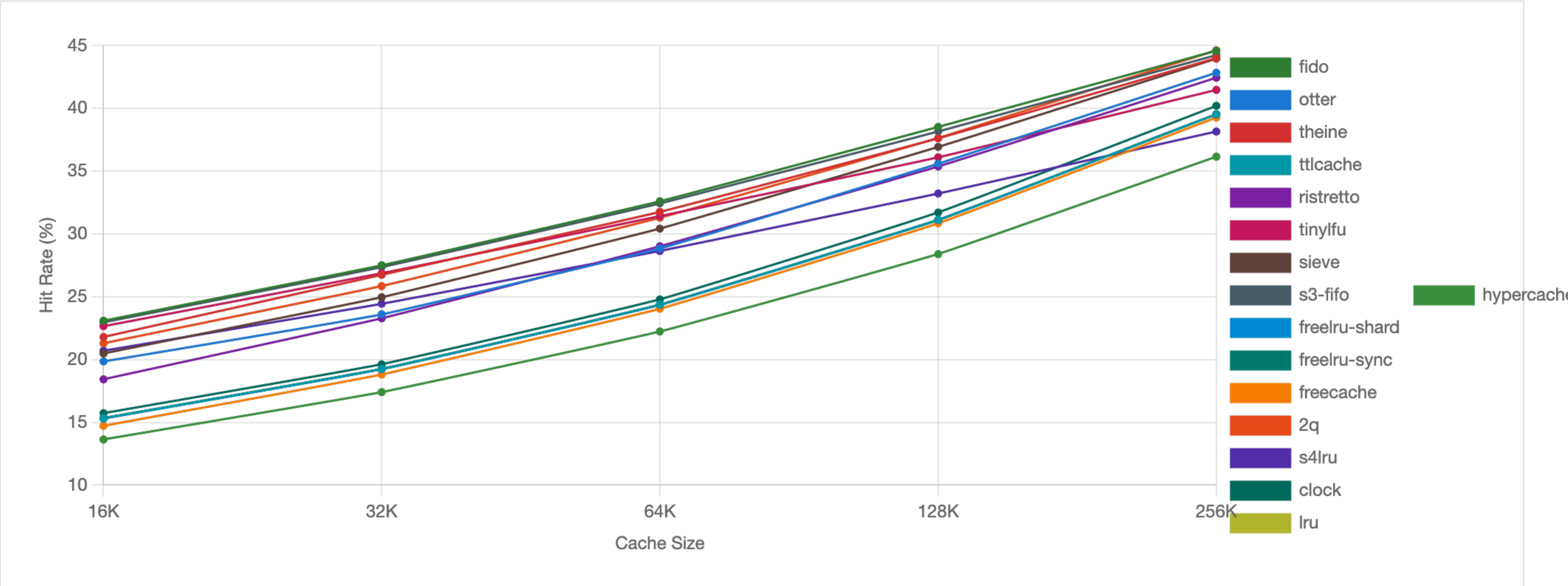
Results Table

Cache	16K	32K	64K	128K	256K	Avg ▼
fido	81.946%	83.717%	84.991%	85.969%	86.773%	84.679%
2q	80.970%	83.050%	84.875%	86.166%	86.773%	84.367%
sieve	80.971%	82.965%	84.599%	86.077%	86.773%	84.277%
s3-fifo	80.891%	82.933%	84.684%	85.969%	86.773%	84.250%
freecache	80.357%	82.578%	84.708%	86.122%	86.773%	84.108%
freelru-sync	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%
lru	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%
ttlcache	80.371%	82.561%	84.500%	86.182%	86.773%	84.077%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	80.341%	82.561%	84.492%	86.179%	86.772%	84.069%
s4lru	80.883%	82.760%	84.250%	85.464%	86.425%	83.957%
clock	80.030%	82.285%	84.257%	85.997%	86.773%	83.868%
otter	79.785%	81.369%	83.918%	85.774%	86.773%	83.524%
tinylfu	78.260%	81.144%	84.030%	85.820%	86.772%	83.205%
hypercache	78.974%	81.245%	83.314%	85.275%	86.773%	83.116%
theine	80.227%	81.524%	82.472%	84.111%	86.771%	83.021%
ristretto	70.413%	72.873%	75.331%	78.890%	85.266%	76.555%

Wikipedia CDN Upload Trace

> 2M operations, upload.wikimedia.org. Source: [libCacheSim]



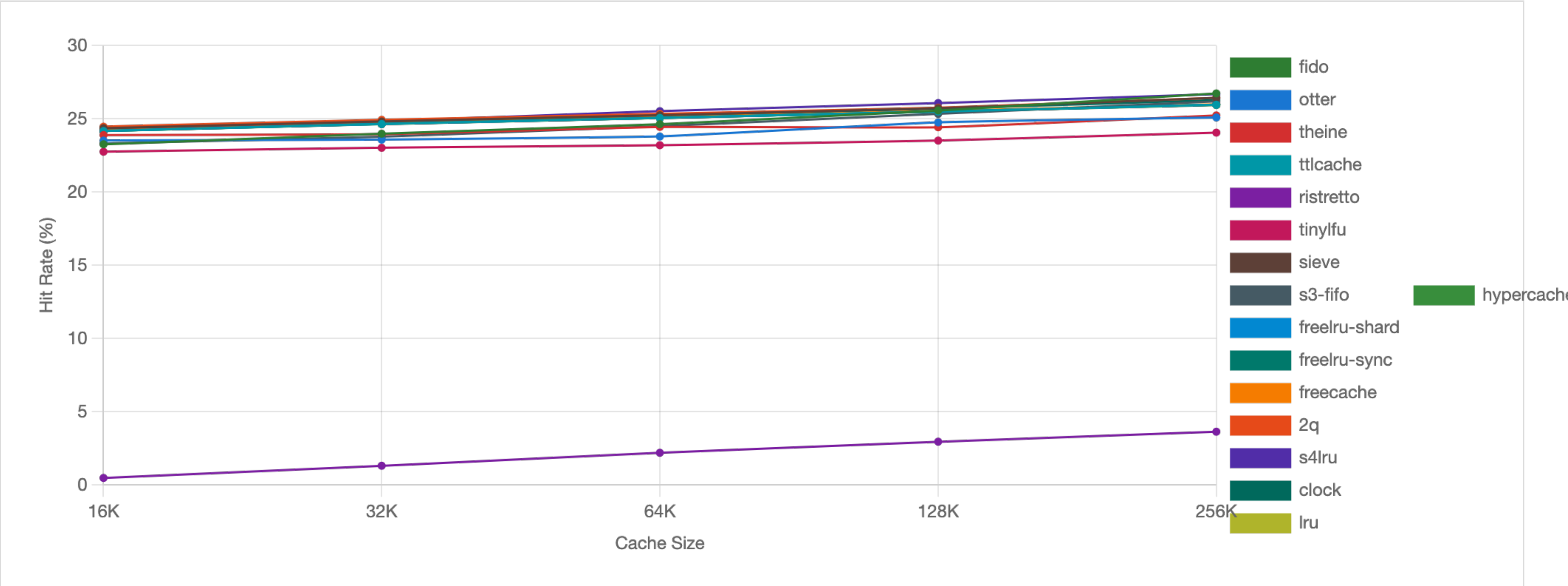
Results Table

Cache	16K	32K	64K	128K	256K	Avg ▼
fido	23.093%	27.506%	32.610%	38.534%	44.601%	33.269%
s3-fifo	22.970%	27.374%	32.441%	38.173%	44.250%	33.042%
theine	21.796%	26.752%	31.762%	37.618%	44.009%	32.387%
2q	21.286%	25.850%	31.285%	37.635%	44.619%	32.135%
tinylfu	22.663%	26.880%	31.434%	36.112%	41.483%	31.714%
sieve	20.476%	24.956%	30.430%	36.932%	43.963%	31.351%
otter	19.847%	23.597%	28.854%	35.597%	42.848%	30.149%
ristretto	18.420%	23.282%	29.023%	35.370%	42.441%	29.707%

Cache	16K	32K	64K	128K	256K	Avg
s4lru	20.697%	24.438%	28.654%	33.215%	38.155%	29.032%
clock	15.725%	19.617%	24.792%	31.711%	40.196%	26.408%
freelru-sync	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
ttlcache	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
lru	15.327%	19.249%	24.375%	31.121%	39.530%	25.920%
freelru-shard	15.299%	19.233%	24.364%	31.117%	39.529%	25.908%
freecache	14.725%	18.808%	24.046%	30.846%	39.278%	25.541%
hypercache	13.633%	17.398%	22.230%	28.402%	36.146%	23.562%

Google Thesios I/O Block Trace

> 400K read operations, ~322K unique blocks. Source: [Tectonic-Shift (SOSP '24)]



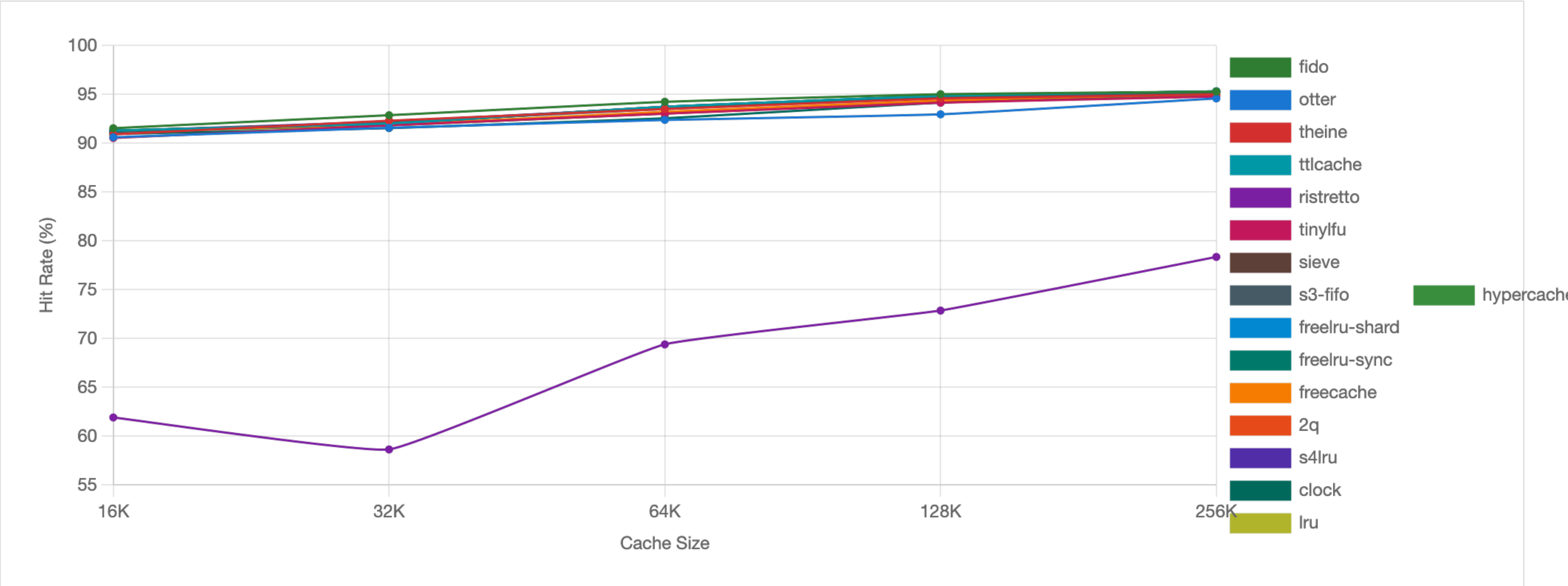
Results Table

Cache	16K	32K	64K	128K	256K	Avg
s4lru	24.191%	24.839%	25.526%	26.082%	26.693%	25.466%
2q	24.467%	24.925%	25.340%	25.760%	26.430%	25.384%
sieve	24.362%	24.818%	25.246%	25.726%	26.428%	25.316%
clock	24.278%	24.756%	25.200%	25.694%	26.280%	25.242%
freecache	24.211%	24.685%	25.103%	25.537%	25.990%	25.105%
freelru-sync	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%
lru	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%
ttlcache	24.184%	24.656%	25.067%	25.501%	25.966%	25.075%

Cache	16K	32K	64K	128K	256K	Avg
freelru-shard	24.177%	24.653%	25.067%	25.501%	25.965%	25.072%
hypercache	24.168%	24.640%	25.059%	25.492%	25.940%	25.060%
fido	23.262%	23.995%	24.636%	25.573%	26.738%	24.841%
s3-fifo	23.308%	23.784%	24.505%	25.336%	26.186%	24.624%
theine	23.899%	23.964%	24.446%	24.433%	25.235%	24.395%
otter	23.516%	23.585%	23.802%	24.756%	25.085%	24.149%
tinylfu	22.756%	23.028%	23.198%	23.519%	24.064%	23.313%
ristretto	0.494%	1.324%	2.206%	2.958%	3.654%	2.127%

Google Thesios I/O File Trace

> 400K read operations, ~46K unique files. Source: [Tectonic-Shift (SOSP '24)]



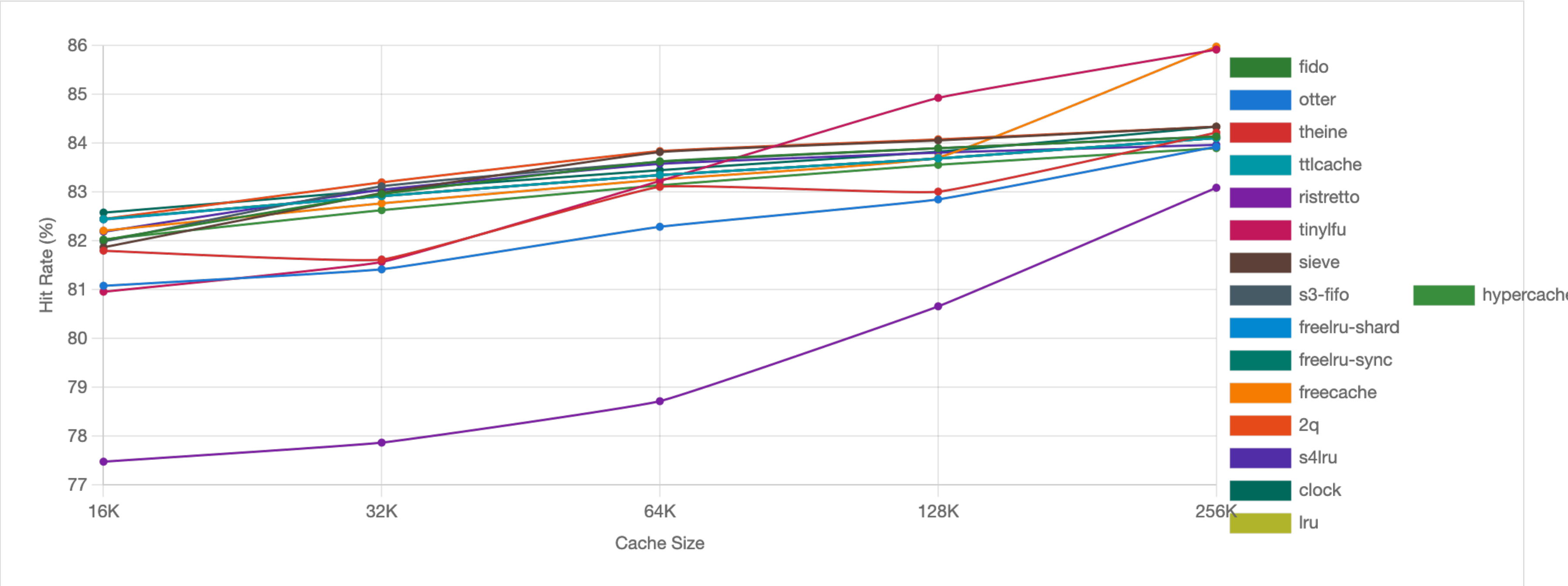
Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	91.553%	92.881%	94.249%	95.030%	95.302%	93.803%
s4lru	91.260%	92.211%	93.663%	94.840%	95.310%	93.457%
lru	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
ttlcache	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
freelru-sync	91.295%	92.027%	93.732%	94.855%	95.311%	93.444%
freelru-shard	91.291%	92.026%	93.730%	94.849%	95.310%	93.441%
sieve	91.287%	92.041%	93.699%	94.840%	95.260%	93.425%
2q	91.279%	92.094%	93.689%	94.714%	95.290%	93.413%

Cache	16K	32K	64K	128K	256K	Avg
s3-fifo	91.193%	92.181%	93.520%	94.671%	95.211%	93.355%
theine	90.920%	92.323%	93.512%	94.589%	94.996%	93.268%
freecache	91.228%	91.898%	93.212%	94.359%	95.056%	93.150%
hypercache	91.190%	91.827%	93.043%	94.214%	94.912%	93.037%
clock	90.965%	91.559%	92.565%	94.175%	95.170%	92.887%
tinylfu	90.536%	91.832%	93.032%	94.147%	94.779%	92.865%
otter	90.615%	91.590%	92.388%	92.970%	94.600%	92.432%
ristretto	61.920%	58.650%	69.410%	72.869%	78.356%	68.241%

IBM Docker Registry Trace

> 725K GET requests, ~121K unique URIs. Source: FAST '18



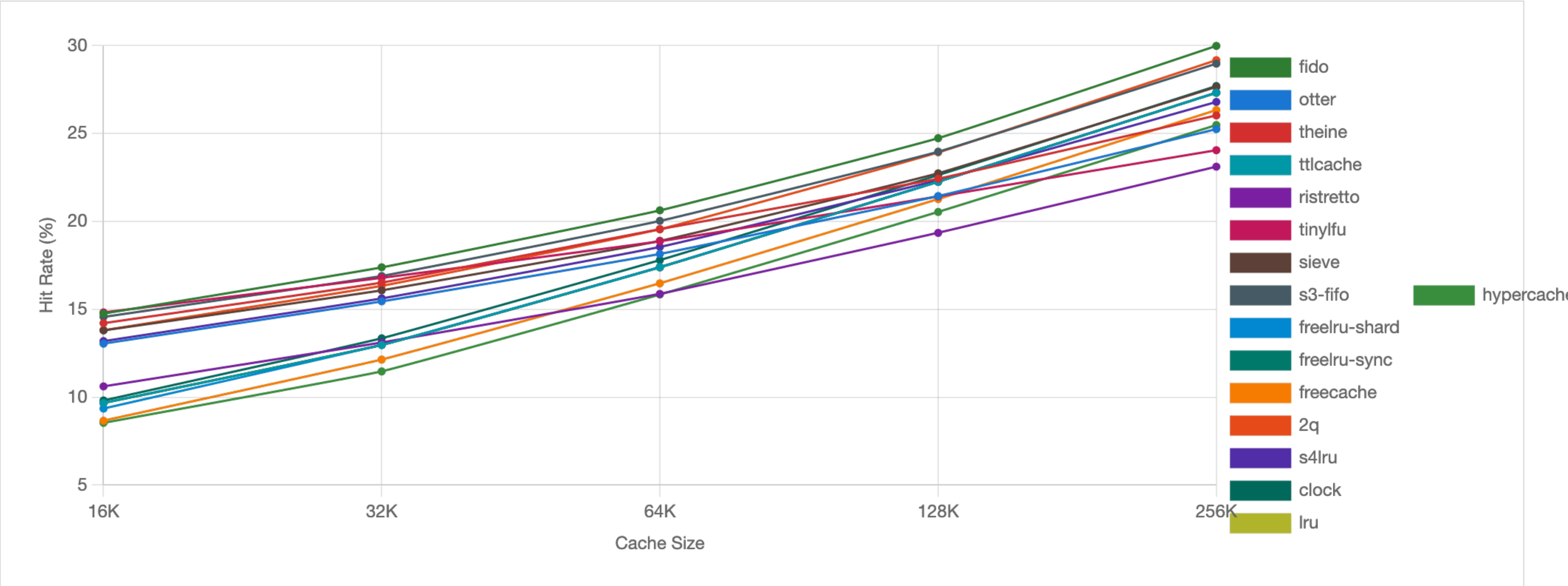
Results Table

Cache	16K	32K	64K	128K	256K	Avg
freecache	82.211%	82.773%	83.263%	83.705%	85.983%	83.587%
2q	82.451%	83.197%	83.840%	84.082%	84.344%	83.583%
clock	82.576%	83.041%	83.446%	83.826%	84.344%	83.446%
sieve	81.868%	82.980%	83.822%	84.064%	84.344%	83.416%
s3-fifo	81.994%	83.124%	83.616%	83.897%	84.141%	83.354%
fido	82.017%	82.962%	83.627%	83.904%	84.141%	83.330%
s4lru	82.189%	83.050%	83.579%	83.810%	83.975%	83.321%
tinylfu	80.956%	81.574%	83.225%	84.927%	85.916%	83.320%

Cache	16K	32K	64K	128K	256K	Avg
lru	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
ttlcache	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
freelru-sync	82.451%	82.922%	83.346%	83.688%	84.110%	83.303%
freelru-shard	82.439%	82.920%	83.348%	83.690%	84.110%	83.301%
hypercache	82.026%	82.626%	83.140%	83.559%	83.896%	83.050%
theine	81.804%	81.625%	83.108%	83.013%	84.217%	82.753%
otter	81.082%	81.418%	82.286%	82.847%	83.933%	82.313%
ristretto	77.481%	77.874%	78.723%	80.663%	83.092%	79.566%

Tencent Photo Trace

> 2M requests, ~1.34M unique photos. Source: ATC '18



Results Table

Cache	16K	32K	64K	128K	256K	Avg
fido	14.755%	17.389%	20.625%	24.738%	29.988%	21.499%
s3-fifo	14.570%	16.903%	20.031%	23.965%	28.982%	20.890%
2q	13.813%	16.336%	19.564%	23.928%	29.182%	20.565%
sieve	13.795%	16.084%	18.895%	22.736%	27.648%	19.831%
theine	14.213%	16.522%	19.556%	22.429%	26.025%	19.749%
s4lru	13.187%	15.624%	18.551%	22.347%	26.802%	19.302%
tinylfu	14.817%	16.786%	18.872%	21.420%	24.051%	19.189%
otter	13.049%	15.452%	18.147%	21.443%	25.248%	18.668%

Cache	16K	32K	64K	128K	256K	Avg
clock	9.811%	13.352%	17.799%	22.631%	27.696%	18.258%
ttlcache	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
lru	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
freelru-sync	9.675%	12.972%	17.390%	22.264%	27.319%	17.924%
freelru-shard	9.351%	12.978%	17.399%	22.254%	27.318%	17.860%
freecache	8.662%	12.138%	16.481%	21.273%	26.328%	16.976%
ristretto	10.609%	13.108%	15.878%	19.351%	23.108%	16.411%
hypercache	8.539%	11.470%	15.835%	20.541%	25.476%	16.372%

Latency Benchmarks (Single-Threaded) ▼

> Lower latency means faster operations. Measures the time (in nanoseconds) for individual Get, Set, and SetEvict operations in a single-threaded environment.

String Keys

Results

Cache	Get (ns)	Get allocs	Set (ns)	Set allocs	SetEvict (ns)	SetEvict allocs	Avg (ns)

Cache	Get (ns)	Get allocs	Set (ns)	Set allocs	SetEvict (ns)	SetEvict allocs	Avg (ns)
fido	10.0	0	16.0	0	119.0	1	13.000
clock	16.0	0	23.0	0	104.0	2	19.500
s4lru	19.0	0	61.0	1	63.0	1	40.000
freelru-shard	24.0	0	38.0	0	46.0	0	31.000
sieve	24.0	0	47.0	0	195.0	3	35.500
freelru-sync	24.0	0	24.0	0	38.0	0	24.000
lru	28.0	0	30.0	0	96.0	1	29.000
2q	29.0	0	31.0	0	247.0	2	30.000
s3-fifo	31.0	0	48.0	0	352.0	5	39.500
otter	35.0	0	137.0	1	167.0	1	86.000
ristretto	59.0	1	266.0	3	115.0	4	162.500
tinylfu	84.0	0	113.0	3	119.0	3	98.500
theine	86.0	1	393.0	0	502.0	1	239.500
freecache	91.0	1	62.0	0	141.0	0	76.500
hypercache	110.0	1	219.0	2	951.0	6	164.500
ttlcache	487.0	0	443.0	0	373.0	3	465.000

Int Keys

Results

Cache	Get (ns)	Get allocs	Set (ns)	Set allocs	SetEvict (ns)	SetEvict allocs	Avg (ns)
freelru-shard	7.0	0	18.0	0	18.0	0	12.500
fido	8.0	0	14.0	0	100.0	1	11.000
otter	32.0	0	136.0	1	157.0	1	84.000
theine	81.0	1	306.0	0	411.0	1	193.500
ttlcache	443.0	0	427.0	0	356.0	3	435.000

String Keys - GetOrSet

> GetOrSet is an atomic operation that gets a value if it exists, or sets it if it doesn't. Only caches that support this operation are shown.

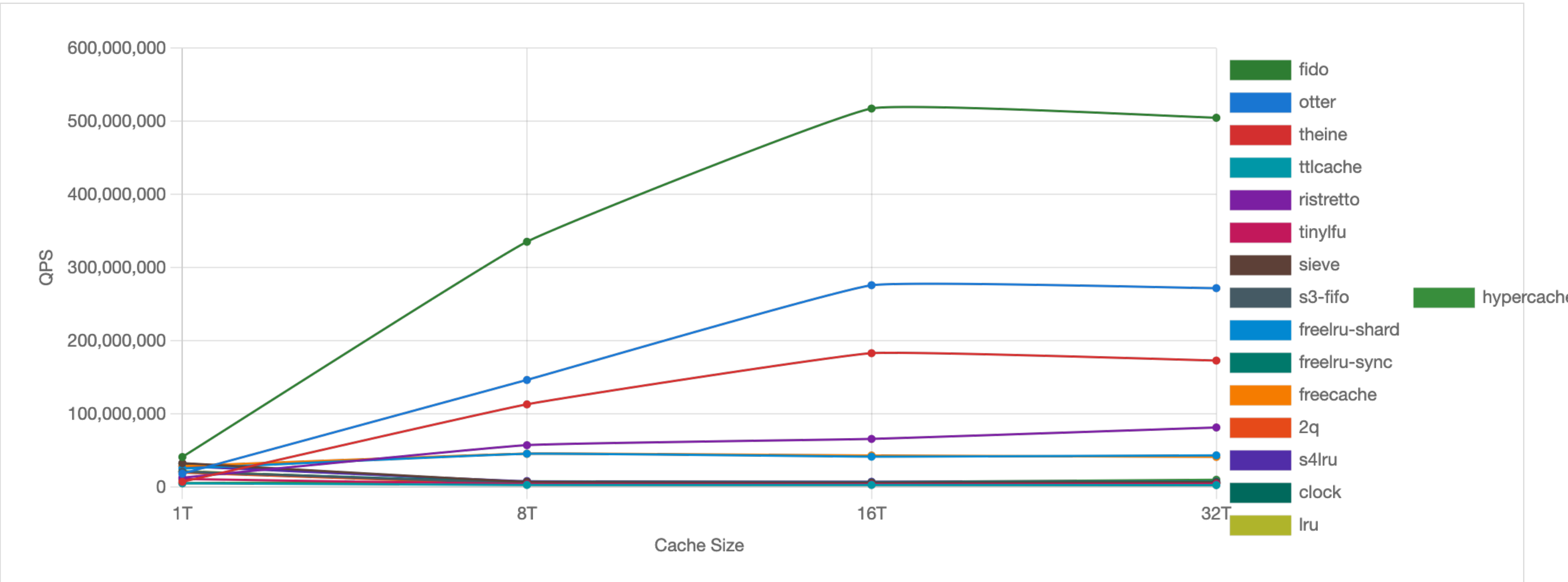
GetOrSet Results

Cache	GetOrSet (ns)	GetOrSet allocs
fido	11.0	0
otter	54.0	1
freecache	127.0	2
theine	165.0	4
hypercache	205.0	2
ttlcache	478.0	0

Throughput Benchmarks (Multi-Threaded) ▼

> Higher throughput means more queries per second (QPS). Tests measure concurrent Get and Set performance separately with a Zipf workload across different thread counts.

String Keys - Get

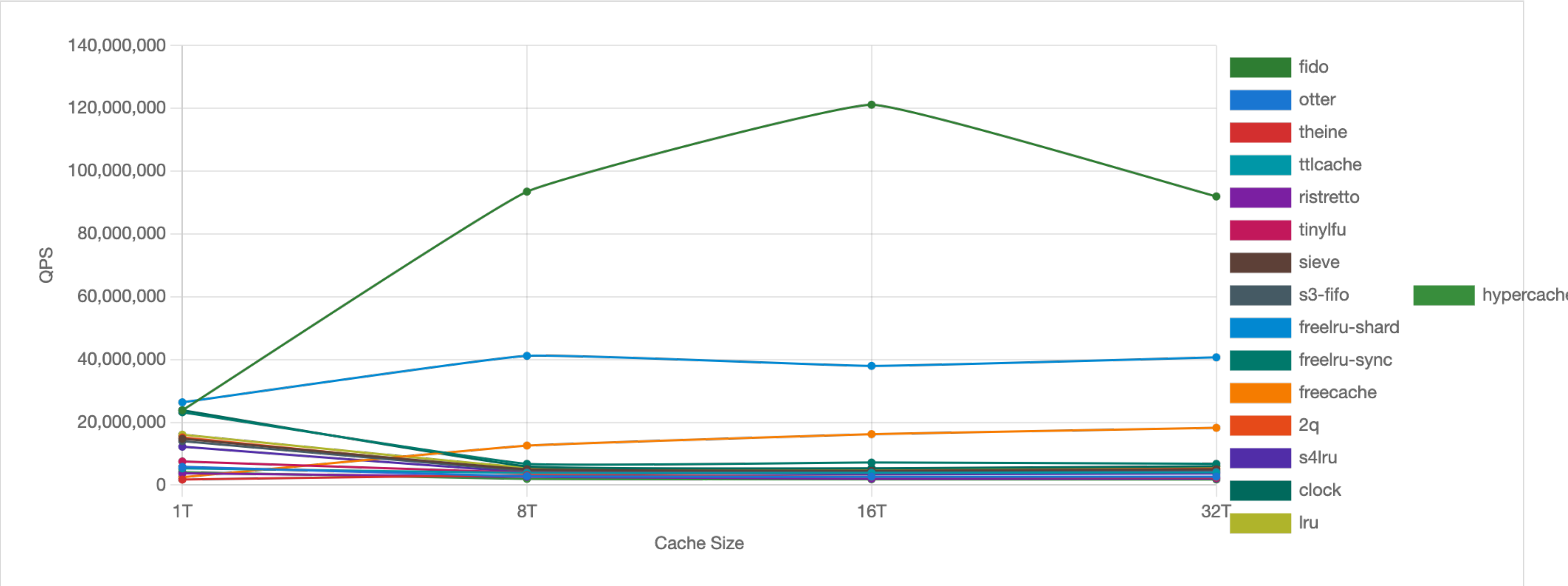


Results Table

Cache	1T	8T	16T	32T	Avg ▼
fido	41.33M	335.53M	517.65M	504.92M	349.86M

Cache	1T	8T	16T	32T	Avg
otter	18.70M	146.74M	276.04M	272.03M	178.38M
theine	7.49M	113.40M	183.26M	173.16M	119.33M
ristretto	13.03M	57.55M	66.07M	81.66M	54.58M
freecache	29.00M	45.90M	43.39M	41.60M	39.97M
freelru-shard	25.55M	45.83M	41.90M	43.57M	39.21M
sieve	32.96M	7.44M	6.14M	6.78M	13.33M
s4lru	28.81M	8.36M	7.57M	7.76M	13.13M
clock	32.41M	6.22M	5.55M	6.02M	12.55M
freelru-sync	21.27M	6.80M	6.42M	7.58M	10.52M
lru	22.09M	5.40M	5.13M	5.52M	9.54M
s3-fifo	21.46M	5.80M	5.21M	5.66M	9.53M
2q	20.17M	4.61M	4.66M	5.65M	8.77M
hypercache	5.62M	8.04M	7.26M	10.20M	7.78M
tinylfu	11.14M	4.42M	4.03M	4.58M	6.04M
ttlcache	5.48M	3.13M	3.09M	2.92M	3.65M

String Keys - Set

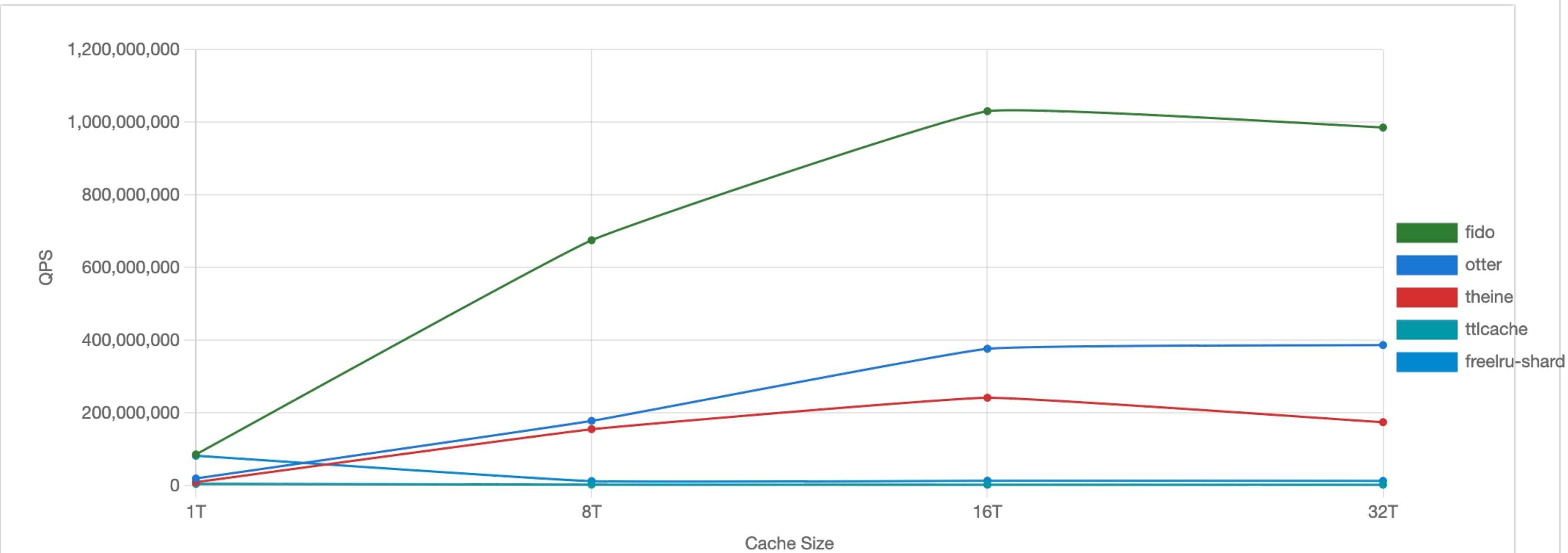


Results Table

Cache	1T	8T	16T	32T	Avg
fido	23.83M	93.49M	121.17M	91.95M	82.61M
freelru-shard	26.35M	41.16M	37.95M	40.68M	36.54M
freecache	2.54M	12.57M	16.23M	18.26M	12.40M
freelru-sync	23.17M	6.75M	7.17M	6.77M	10.97M
clock	23.86M	5.99M	5.37M	6.03M	10.31M
lru	16.07M	5.61M	4.90M	5.85M	8.11M
2q	15.18M	4.67M	4.41M	5.25M	7.38M
sieve	14.74M	5.04M	4.54M	5.02M	7.34M

Cache	1T	8T	16T	32T	Avg
s3-fifo	13.98M	4.64M	4.13M	4.61M	6.84M
s4lru	12.26M	4.23M	3.50M	3.70M	5.92M
tinylfu	7.54M	4.04M	3.85M	4.17M	4.90M
ttlcache	5.36M	3.90M	3.92M	4.09M	4.32M
otter	5.83M	2.86M	2.66M	2.68M	3.51M
ristretto	3.76M	2.66M	2.04M	2.08M	2.63M
theine	1.76M	3.16M	2.63M	2.42M	2.49M
hypercache	4.11M	1.99M	1.84M	1.80M	2.44M

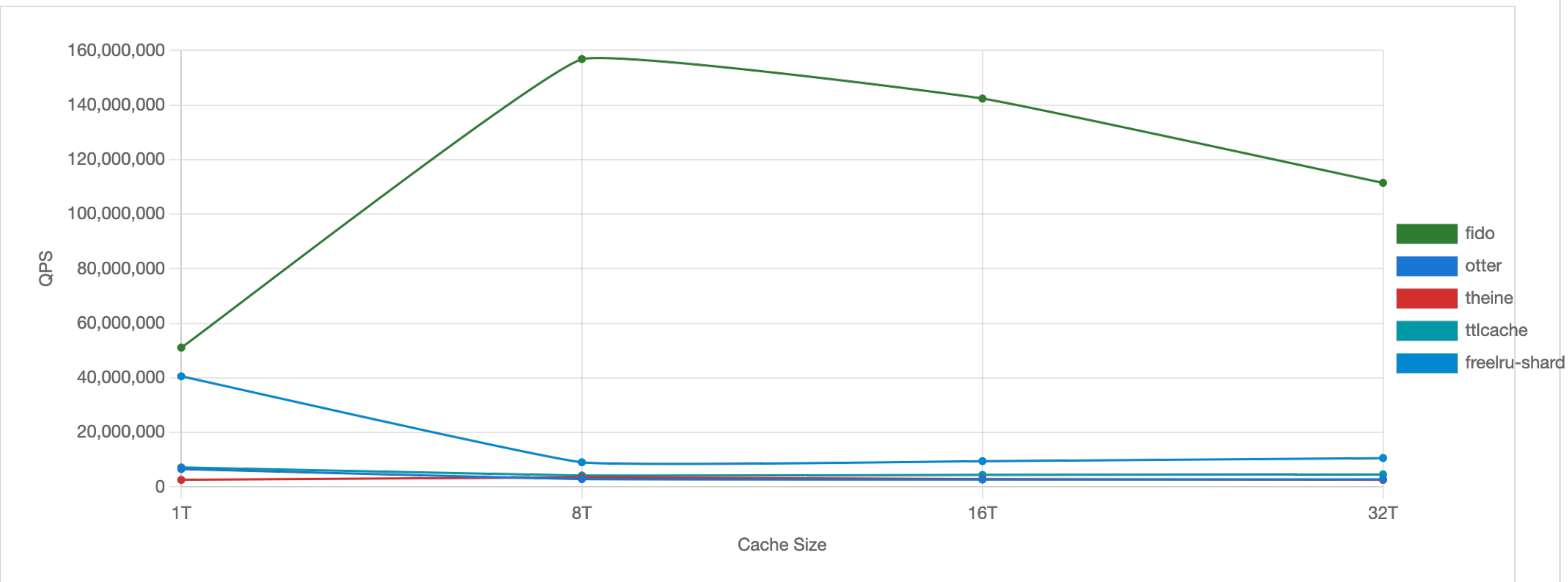
Int Keys - Get



Results Table

Cache	1T	8T	16T	32T	Avg
fido	86.03M	675.47M	1030.41M	985.70M	694.40M
otter	20.16M	178.60M	376.86M	387.22M	240.71M
theine	10.14M	155.63M	242.47M	175.04M	145.82M
freelru-shard	82.82M	12.82M	13.81M	13.55M	30.75M
ttlcache	5.46M	3.26M	3.08M	2.89M	3.67M

Int Keys - Set

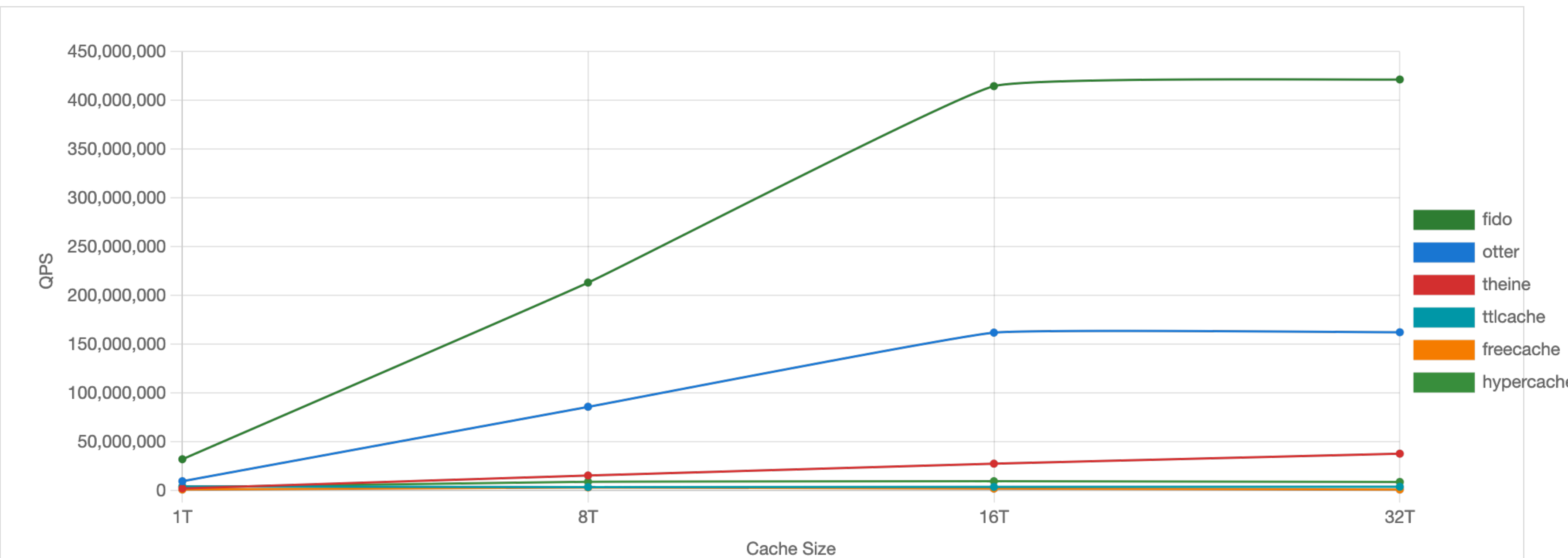


Results Table

Cache	1T	8T	16T	32T	Avg
fido	51.09M	156.86M	142.42M	111.48M	115.46M
freelru-shard	40.60M	9.07M	9.48M	10.59M	17.44M
ttlcache	7.19M	4.24M	4.44M	4.59M	5.12M
otter	6.57M	2.89M	2.71M	2.80M	3.74M
theine	2.60M	3.50M	2.87M	2.61M	2.89M

GetOrSet Operations

> Atomic get-or-compute operations with URL-like keys. Only caches supporting GetOrSet are shown.



Results Table

Cache	1T	8T	16T	32T	Avg ▼
fido	32.33M	213.41M	414.75M	421.54M	270.51M
otter	9.77M	86.01M	162.10M	162.49M	105.09M
theine	2.34M	15.59M	27.75M	38.01M	20.92M
hypercache	3.62M	9.23M	9.84M	9.08M	7.94M
ttlcache	4.47M	3.73M	3.95M	4.11M	4.07M
freecache	1.30M	3.27M	2.26M	1.21M	2.01M

Memory Overhead Benchmarks

> Capacity: 32768 items, Value size: 1024 bytes. Measured in isolated processes. Overhead compared to baseline `map[string][]byte`. Lower is better.

Memory Results

Cache	Items Stored	Memory (MB)	Overhead vs map (bytes/item)
freelru-sync	32768	35.32	-20
otter	32768	36.45	15

Cache	Items Stored	Memory (MB)	Overhead vs map (bytes/item)
freelru-shard	31606	36.60	20
clock	32768	37.13	37
fido	32768	37.52	49
lru	32768	37.65	53
2q	32768	37.65	53
s4lru	32768	38.40	78
theine	32768	38.61	84
sieve	32768	39.66	118
s3-fifo	32768	39.66	118
tinylfu	32767	39.97	128
ttlcache	32768	40.94	159
ristretto	32768	41.59	179
hypercache	32768	43.39	237
freecache	32764	44.90	285